

aCount

Sneaker Resale Accounting & Financial Management

Technical Specification Document

Version 1.0 — March 2026

Prepared by: Gregorio Delfin P. Pascua

1. Project Overview

aCount is a personal web application for accounting and financial record management tailored for sneaker resellers. It automates financial tracking through Gmail integration by collecting e-receipts, order confirmations, sales invoices, and bank transfer notifications. The primary sales platform is Alias (alias.org).

Primary User: Sneaker reseller operating in the Philippines

Primary Sales Platform: Alias (uses USD for transactions)

Base Currency: Philippine Peso (PHP) — all final accounting outputs in PHP

Scope: Single-user application (personal use)

2. Core Features

2.1 Sales Tracking

Track the full order lifecycle on Alias from initial sale notification through completion or cancellation. Each sale record must capture order details, status transitions, timestamps, and financial outcomes.

2.1.1 Order Lifecycle States

Status	Trigger	Description
Pending	"Just Sold" email received	Sale detected; awaiting seller confirmation within 24 hours
Confirmed	"Shipping Label" email received	Seller confirmed; shipping label generated with deadline
Shipped	"Shipped to alias" email received	Package picked up and in transit to Alias for verification
Completed	"Completed" email received	Item authenticated; earnings available for cash out
Cancelled	Cancellation email received	Order cancelled (multiple sub-reasons — see section 2.1.2)
Attention Needed	"Attention Needed" email received	Authentication issue found; discount negotiation or action required
Consigned	Seller selects consignment option	Item held by Alias for relisting (90-day window, then \$2/month storage fee)
Returned	Seller requests return	Item shipped back; retrieval fee charged

2.1.2 Seller Cancellation Types

Cancellations initiated by the seller (as opposed to Attention Needed cancellations triggered by authentication issues) fall into two distinct categories based on whether the order was confirmed before cancellation:

Cancellation Type	Condition	Fee	Email Indicators
Unconfirmed Cancellation	Sale was received but seller did NOT confirm within 24 hours, or seller actively cancelled before confirming	No fee charged	Body contains: "Your order has been canceled and your listing has been removed." Does NOT mention a fee or charge amount.
Confirmed Cancellation	Sale was confirmed (shipping label generated) but seller failed to ship within the 3-day window, or seller actively cancelled after confirming	\$10 fee charged	Body contains: "Because this was a confirmed order, you will be charged \$10." May also mention the 3-day shipping window.

Both cancellation email types share the same subject pattern: "Order #XXXXXX Has Been Canceled" from info@alias.org. The system must parse the email body to distinguish between the two types and determine whether a fee applies.

Key parsing logic: If the email body contains the phrase "you will be charged \$10" (or any dollar amount), classify as Confirmed Cancellation and create a corresponding Fee expense record. Otherwise, classify as Unconfirmed Cancellation with no fee.

Important note from Alias: Multiple cancellations result in a lower seller score and higher commission rates. The system should track cancellation frequency as a seller health metric.

2.1.3 Attention Needed Flow (Detailed)

This is the most complex workflow. When Alias authentication finds an issue, the order enters a branching decision tree:

Issue Type	System Behavior	Possible Outcomes
Used Product / Authentication Fail	Seller prompted to offer discount; auto 5% discount after 48hrs if no response	Buyer accepts discount → sale completes at reduced price; Buyer declines → cancellation (see below)
Wrong Size or SKU	Automatic consignment and cancellation of original order	Item consigned under new listing; original sale cancelled

Issue Type	System Behavior	Possible Outcomes
Mold	Item returned immediately with a fee	Fee charged; item shipped back to seller; sale cancelled

When a buyer declines the discount:

1. Original order is cancelled.
2. A NEW order number is generated (original order number + 1) for the consignment or return process.
3. Seller chooses Option A (Consign) or Option B (Return with retrieval fee).
4. The system must track the relationship between original and new order numbers.

State Diagram: Attention Needed → Discount Offered → Buyer Accepts (Completed at discount) | Buyer Declines → New Order# → Consign | Return

2.1.4 Shipment Deadline Tracking

Each confirmed order includes a shipment deadline (parsed from the confirmation email). The system must:

- Display days remaining until shipment deadline on the dashboard
- Alert the user when a deadline is within 24 hours
- Flag overdue shipments as critical action items
- Track the scheduled pickup window (date, time range, address)

2.1.5 Profit Calculation Logic

Alias reports earnings in USD. However, true profit must be calculated in PHP based on actual bank transfers received:

- Revenue (PHP): Sourced from Bank Transfer Completed emails (the actual PHP amount deposited)
- Cost (PHP): Sourced from Inventory PurchaseCost field
- Gross Profit = Bank Transfer Amount (PHP) – Purchase Cost (PHP)
- Net Profit = Gross Profit – Fees (retrieval, storage, cancellation) – Associated Expenses

Important: The USD "Amount Made" from Alias emails is informational only. The PHP bank transfer amount is the source of truth for revenue, as it reflects actual conversion rates and any Alias-side deductions.

2.2 Inventory Management

Maintain a real-time database of owned sneakers with individual item tracking. Inventory should support both automated and manual entry.

2.2.1 Inventory Population Methods

Method	Trigger	Details
Automated (Gmail)	Purchase confirmation email detected	Parse SKU, shoe name, size, price, date, source from email
Manual Entry	User adds item via UI	For purchases made outside tracked platforms (in-person, other marketplaces)

2.2.2 Inventory Status Transitions

Event	Status Change	Additional Action
Item purchased / added	N/A → Available	Create new inventory record
Sale detected	Available → Sold	Link to SaleID; match via FIFO (see 2.2.3)
Sale completed	Sold (remains)	No change; already linked
Cancellation / Return	Sold → Available	Unlink SaleID; restore to inventory
Consignment	Sold → Consigned	Track consignment start date for storage fee calculation

2.2.3 FIFO Matching Rules

When a sale is detected, the system matches it to inventory using First-In-First-Out (FIFO) logic:

- Match by SKU + Size + Status = Available.
- If multiple matches exist, select the item with the earliest DatePurchased.
- If no match is found (item not in inventory), create the sale record anyway and flag it with a "No Inventory Match" indicator.
- The flag does NOT imply an error — the shoe may simply not have been manually recorded yet.
- When the user later adds the inventory item, the system should prompt to link it to the unmatched sale.

2.2.4 Inventory Data Fields

See Section 5 (Data Schema) for the full entity definition. Key fields include: SKU, ShoeName, Size, DatePurchased, PurchaseCost (PHP), ListedPrice (USD), Status, and LinkedSaleID.

2.3 Financial Tracking

Beyond sneaker sales, the system must capture all financial activity to provide a complete picture of the user's finances.

2.3.1 Revenue Sources

- Sneaker sales (via Alias bank transfers in PHP)
- Any other income sources the user manually adds

2.3.2 Expense Categories

Category	Source	Examples
Sneaker Purchases	Gmail / Manual	Cost of goods for resale
Platform Fees	Gmail (Alias emails)	Cancellation fees, retrieval fees, storage fees
Subscriptions	Gmail / Manual	Netflix, Grab, Spotify, etc.
Personal Online Orders	Gmail (Shopee, Lazada, etc.)	Non-resale personal purchases
Other Expenses	Manual Entry	Shipping supplies, travel, etc.

2.3.3 Subscription Tracking

The system should maintain a dedicated subscription tracker with the following attributes:

- Subscription Name (e.g., Netflix, Grab, Spotify)
- Amount and Currency (with PHP conversion)
- Billing Cycle (monthly, annual, etc.)
- Next Billing Date (auto-calculated or parsed from emails)
- Status (Active, Paused, Cancelled)
- Payment Method reference

2.3.4 Bank Transfer Tracking

Each "Bank Transfer Completed" email from Alias must be parsed and stored as an independent financial event:

- Transfer Amount (PHP — already in PHP from the email)
- Bank Account (last 4 digits + bank name from email)
- Transfer Date and Time
- Reconciliation Status: link the transfer to one or more completed sale(s)

Note: A single bank transfer may cover earnings from multiple completed sales. The system should support allocating one transfer across multiple SaleIDs, or flagging unreconciled transfers for manual review.

2.4 Currency Conversion

All Alias transactions are in USD, but the application's base currency is PHP. The system needs a defined conversion strategy:

Approach	When to Use	Implementation
Live API Rate	Recording new expenses from non-PHP sources	Call a currency conversion API (e.g., exchangeratesapi.io , Open Exchange Rates) at time of record creation
Actual Bank Transfer	Calculating true sales revenue	Use the PHP amount from the Bank Transfer Completed email — this reflects the real conversion rate Alias applied
Manual Override	User wants to set a specific rate	Allow manual rate entry for individual transactions

The USD values from Alias (SellingPrice, AmountMade) should be stored as-is for reference, but all financial reports, dashboards, and profit calculations must use PHP values.

3. Gmail Integration

3.1 Email Monitoring Scope

The application monitors the user's Gmail inbox for specific email patterns. Authentication is handled via Gmail API OAuth 2.0 with a persistent refresh token so the user does not need to re-authenticate frequently.

3.2 Email Classification Rules

Emails are classified by sender address and subject line pattern. The following table defines the primary classification rules:

Email Type	Sender	Subject Pattern	System Action
Sale Notification	info@alias.org	"Your [Name] Just Sold" or contains "Just Sold"	Create Sale record (Pending); attempt FIFO inventory match
Order Confirmed	info@alias.org	"Order #XXXXXXX - Shipping Label and Instructions"	Update sale to Confirmed; parse shipment deadline and pickup details
Shipped	info@alias.org	"Order #XXXXXXX - Shipped to alias for Verification"	Update sale to Shipped

Email Type	Sender	Subject Pattern	System Action
Completed	info@alias.org	"Order #XXXXXX Completed" and contains "Available for Cash Out"	Update sale to Completed; record USD earnings amount
Attention Needed	info@alias.org	"Order #XXXXXX - Attention Needed"	Update sale to Attention Needed; parse issue type
Buyer Accepted	info@alias.org	"Order #XXXXXX - Buyer Accepted"	Update sale back to processing; await completion email
Cancellation (No Fee)	info@alias.org	"Order #XXXXXX Has Been Canceled" — body does NOT mention a charge	Update sale to Cancelled; restore inventory; no fee record
Cancellation (With Fee)	info@alias.org	"Order #XXXXXX Has Been Canceled" — body contains "you will be charged \$XX"	Update sale to Cancelled; restore inventory; create \$10 Fee expense record
Bank Transfer	info@alias.org	"Bank Transfer Completed"	Create BankTransfer record; parse PHP amount, bank details, date
Purchase Confirmation	Various	Contains order confirmation keywords	Create Inventory record; parse item details
Subscription Charge	Various	Billing or receipt from known subscription services	Create/update Expense record under Subscriptions
Personal E-Receipt	Various (Shopee, Lazada, etc.)	Order confirmation or receipt	Create Expense record under Personal Orders

3.3 Email Parsing Specifications

For Alias emails (info@alias.org), the following data fields must be extracted:

3.3.1 Sale Notification ("Just Sold")

Field	Extraction Pattern	Example Value
Order Number	"Order #XXXXXX" from subject/body	992603383
Shoe Name	Line after "Name:" or from subject line	Nike Wmns Air Rift 'Healing Orange'
Condition	Line after "Condition:"	New
Box Condition	Line after "Box:"	Good Condition
Size	Line after "Size:"	8
SKU	Line after "SKU:"	HM8288 800
Selling Price (USD)	Line after "Price:"	\$72

Field	Extraction Pattern	Example Value
Amount Made (USD)	Line after "Amount Made:"	\$53.07

3.3.2 Confirmation Email

Field	Extraction Pattern	Example Value
Shipment Deadline	"must be shipped by [DATE TIME TIMEZONE]"	March 10, 2026 00:00 Asia/Manila
Pickup Address	Lines after "ready for Janio collection at:"	Block 1 Unit 27, Romarosa Town Homes...
Pickup Window	"on [DATE] between [TIME] and [TIME]"	2026-03-07 between 9 AM and 7 PM
Earnings (USD)	"Your Earnings: \$XX.XX"	\$53.07

3.3.3 Bank Transfer Email

Field	Extraction Pattern	Example Value
Amount (PHP)	"direct deposit of ₱XX,XXX.XX"	₱45,417.60
Bank Name	"bank account ending in **BANK_NAME"	BANCO DE ORO UNIVERSAL BANK
Account (Last 4)	Digits after bank name, before "has cleared"	7425
Transfer Date	"completed on [DATE TIME]"	March 06, 2026 18:52

3.3.4 Attention Needed Email

Field	Extraction Pattern	Example Value
Order Number	"Order #XXXXXX" from subject	236989906
Issue Type	Text after "discovered the following issue(s):"	used product
Auto-Discount Timer	"5% discount...within 48 hours" (fixed rule)	48 hours from email timestamp

3.3.5 Cancellation Email ("Has Been Canceled")

Both cancellation types use the same subject pattern. The body must be parsed to determine which type:

Field	Extraction Pattern	Example Value
Order Number	"Order #XXXXXX" from subject line	784376048 or 573784020
Cancellation Type	Check body for fee language (see below)	Unconfirmed or Confirmed
Fee Amount	"you will be charged \$XX" — extract dollar amount if present	\$10 (or NULL if no fee)

Detection logic: Scan the email body for the string "you will be charged". If found, extract the dollar amount following it and classify as Confirmed Cancellation. If not found, classify as Unconfirmed Cancellation.

Sample email body (Unconfirmed — No Fee):

"Your order has been canceled and your listing has been removed. Please keep your listings up-to-date to prevent future cancellations. We monitor all seller cancellations to determine if commission rates should be increased or accounts suspended altogether."

Sample email body (Confirmed — With Fee):

"Your order has been canceled and your listing has been removed. Because this was a confirmed order, you will be charged \$10. If you failed to ship your order within the 3-day window, your order was automatically canceled. As a reminder, you have 3 business days to ship an order after it has been confirmed."

3.4 Polling Strategy

The Gmail API integration should use a polling approach (or push notifications via Gmail API watch if feasible) to check for new emails:

- Polling interval: configurable, default every 5 minutes
- Use Gmail API's history.list or messages.list with a "after:" date filter to fetch only new messages
- Track the last-processed email timestamp/historyId to avoid duplicate processing
- Queue emails for processing to handle bursts gracefully

3.5 Authentication

Gmail OAuth 2.0 implementation:

- Scopes required: gmail.readonly (reading emails), gmail.labels (optional, for organizing processed emails)
- Store refresh token securely (encrypted in database or environment variable)
- Auto-refresh access tokens; alert the user if re-authentication is needed
- Single-user: one Gmail account connected at a time

4. Key Workflows

4.1 Purchase Flow

10. Email detected: Purchase confirmation from tracked source
11. Parse: Extract SKU, Shoe Name, Size, Purchase Price, Date, Source
12. Convert: Purchase Price to PHP (if not already PHP) using live API rate
13. Create: New Inventory record with Status = Available
14. Dashboard: Inventory count and value updated in real-time

4.2 Sale Flow

15. Email detected: "Your [Shoe] Just Sold" from info@alias.org
16. Parse: Order#, SKU, Size, Condition, Box Condition, Selling Price (USD), Amount Made (USD)
17. Match: Find Available inventory item using FIFO (SKU + Size, earliest DatePurchased)
18. Create: Sale record with Status = Pending
19. Update Inventory: Status = Sold, LinkedSaleID set (or flag "No Inventory Match" if unmatched)
20. Confirmation email → Update sale to Confirmed, store shipment deadline
21. Shipped email → Update sale to Shipped
22. Completed email → Update sale to Completed
23. Bank Transfer email → Record PHP revenue, reconcile with sale(s)

4.3 Cancellation / Return Flow

Cancellations triggered by the seller follow one of two paths depending on whether the order was confirmed:

Path A — Unconfirmed Cancellation (No Fee):

24. Email detected: "Order #XXXXXXX Has Been Canceled" with no fee language in body
25. Find: Sale record by OrderNumber
26. Update Sale: Status = Cancelled, CancellationDate set, CancellationType = Unconfirmed
27. Restore Inventory: Status = Available, LinkedSaleID = NULL
28. No fee record created

Path B — Confirmed Cancellation (With Fee):

29. Email detected: "Order #XXXXXXX Has Been Canceled" with "you will be charged \$10" in body
30. Find: Sale record by OrderNumber

31. Update Sale: Status = Cancelled, CancellationDate set, CancellationType = Confirmed
32. Create: Fee expense record (\$10 USD, converted to PHP, Category = Platform Fee, linked to SaleID)
33. Restore Inventory: Status = Available, LinkedSaleID = NULL

Path C — Attention Needed Cancellation:

34. Triggered by buyer declining discount (see Section 4.4 Attention Needed Flow)
35. New order number generated (original + 1)
36. Seller chooses Consign or Return — fees vary by option

4.4 Attention Needed Flow

37. Email detected: "Attention Needed" from info@alias.org
38. Parse: Issue type (used product, wrong size/SKU, mold)
39. Branch by issue type:

Path A — Used Product / Authentication Issue:

- Update sale to Attention Needed
- Record 48-hour auto-discount deadline
- If Buyer Accepted email arrives → resume normal flow (awaiting Completed)
- If Buyer Declined → Original order cancelled, new order# = original + 1
- New order enters Consign or Return path

Path B — Wrong Size or SKU:

- Auto-consign item; cancel original sale
- Create consignment record with new order number

Path C — Mold:

- Cancel sale immediately
- Charge fee; create Fee record
- Return item to seller; update inventory status

4.5 Bank Transfer Reconciliation Flow

40. Email detected: "Bank Transfer Completed" from info@alias.org
41. Parse: PHP amount, bank details, date
42. Create: BankTransfer record
43. Auto-reconcile: Match transfer to completed sale(s) by date proximity and amount
44. If auto-match fails: Flag as unreconciled for manual review

45. Update profit calculations using the PHP transfer amount as revenue

5. Data Schema

The following entity definitions cover all data requirements. Primary keys are auto-incrementing integers unless noted otherwise.

5.1 Inventory

Attribute	Type	Description
InventoryID	INT, PK, Auto-increment	Unique identifier
SKU	VARCHAR	Product SKU code (e.g., HM8288 800)
ShoeName	VARCHAR	Full product name
Size	DOUBLE	Shoe size (numeric, supports half sizes)
DatePurchased	DATETIME	Date the item was purchased
PurchaseCost	DECIMAL(10,2)	Cost in PHP
ListedPrice	DECIMAL(10,2)	Listed selling price in USD on Alias
Status	ENUM	Available, Sold, Consigned
LinkedSaleID	INT, FK → Sales	Reference to sale record (NULL if available)
Source	VARCHAR	Where purchased (e.g., Nike PH, Shopee, In-store)
Notes	TEXT	Free-text notes
CreatedAt	DATETIME	Record creation timestamp
UpdatedAt	DATETIME	Last modification timestamp

5.2 Sales

Attribute	Type	Description
SaleID	INT, PK, Auto-increment	Unique identifier
OrderNumber	INT	Alias order number
ParentOrderNumber	INT, NULLABLE	Original order# if this is a discount-declined re-order
Platform	VARCHAR	Sales platform (default: Alias)
SKU	VARCHAR	Product SKU
ShoeName	VARCHAR	Product name
Size	DOUBLE	Shoe size

Attribute	Type	Description
Condition	ENUM	New, Used
BoxCondition	ENUM	Good Condition, No Box, Badly Damaged
SellingPrice	DECIMAL(10,2)	Listed price in USD
AmountMade	DECIMAL(10,2)	Net earnings in USD (after Alias fees)
SaleDate	DATETIME	When the sale was detected
Status	ENUM	Pending, Confirmed, Shipped, Completed, Cancelled, Attention Needed, Consigned, Returned
IssueType	VARCHAR, NULLABLE	Issue description if Attention Needed
DiscountOffered	DECIMAL(10,2), NULLABLE	Discount amount offered to buyer (USD)
ConfirmationDatetime	DATETIME, NULLABLE	When seller confirmed the sale
ShipmentDeadline	DATETIME, NULLABLE	Deadline to ship
PickupAddress	TEXT, NULLABLE	Scheduled pickup address
PickupWindow	VARCHAR, NULLABLE	Pickup date and time range
ShipmentDate	DATETIME, NULLABLE	Actual ship date
CompletionDate	DATE, NULLABLE	When sale completed
CancellationDate	DATE, NULLABLE	When sale was cancelled
CancellationType	ENUM, NULLABLE	Unconfirmed, Confirmed, Attention Needed — reason/type of cancellation
CancellationFee	DECIMAL(10,2), NULLABLE	Fee charged for cancellation in USD (e.g., \$10 for confirmed cancellations)
TrackingNumber	VARCHAR, NULLABLE	Shipment tracking number
InventoryMatchStatus	ENUM	Matched, Unmatched — indicates if FIFO match was found
Notes	TEXT	Free-text notes
CreatedAt	DATETIME	Record creation timestamp
UpdatedAt	DATETIME	Last modification timestamp

5.3 BankTransfers

Attribute	Type	Description
TransferID	INT, PK, Auto-increment	Unique identifier
AmountPHP	DECIMAL(12,2)	Transfer amount in PHP

Attribute	Type	Description
BankName	VARCHAR	Receiving bank name
AccountLast4	VARCHAR(4)	Last 4 digits of bank account
TransferDate	DATETIME	When transfer completed
ReconciliationStatus	ENUM	Reconciled, Partially Reconciled, Unreconciled
Notes	TEXT	Free-text notes
CreatedAt	DATETIME	Record creation timestamp

5.4 BankTransferAllocations (Junction Table)

Links bank transfers to one or more sales (many-to-many relationship for partial/combined transfers):

Attribute	Type	Description
AllocationID	INT, PK, Auto-increment	Unique identifier
TransferID	INT, FK → BankTransfers	The bank transfer
SaleID	INT, FK → Sales	The linked sale
AllocatedAmount	DECIMAL(12,2)	PHP amount allocated to this sale

5.5 Expenses

Attribute	Type	Description
ExpenseID	INT, PK, Auto-increment	Unique identifier
Category	ENUM	Platform Fee, Subscription, Personal Order, Sneaker Purchase, Other
Description	VARCHAR	What the expense is for
AmountOriginal	DECIMAL(10,2)	Amount in original currency
OriginalCurrency	VARCHAR(3)	Currency code (PHP, USD, etc.)
AmountPHP	DECIMAL(10,2)	Converted amount in PHP
ConversionRate	DECIMAL(10,4)	Exchange rate used for conversion
ExpenseDate	DATE	Date of expense
Source	VARCHAR	Email source or Manual
LinkedSaleID	INT, FK, NULLABLE	If this expense is tied to a specific sale (e.g., cancellation fee)

Attribute	Type	Description
Notes	TEXT	Free-text notes
CreatedAt	DATETIME	Record creation timestamp

5.6 Subscriptions

Attribute	Type	Description
SubscriptionID	INT, PK, Auto-increment	Unique identifier
Name	VARCHAR	Service name (Netflix, Grab, Spotify, etc.)
AmountOriginal	DECIMAL(10,2)	Billing amount in original currency
OriginalCurrency	VARCHAR(3)	Currency code
AmountPHP	DECIMAL(10,2)	Billing amount converted to PHP
BillingCycle	ENUM	Monthly, Quarterly, Annual
NextBillingDate	DATE	Next expected charge date
Status	ENUM	Active, Paused, Cancelled
PaymentMethod	VARCHAR	Payment method reference
Notes	TEXT	Free-text notes
CreatedAt	DATETIME	Record creation timestamp
UpdatedAt	DATETIME	Last modification timestamp

5.7 EmailProcessingLog

Track all processed emails to prevent duplicate processing and enable debugging:

Attribute	Type	Description
LogID	INT, PK, Auto-increment	Unique identifier
GmailMessageID	VARCHAR	Gmail's unique message ID
EmailType	ENUM	Sale, Confirmation, Shipped, Completed, Cancelled, Attention, BankTransfer, Purchase, Subscription, Receipt, Other
ProcessedAt	DATETIME	When the email was processed
ParsedData	JSON	Raw parsed data from the email
Status	ENUM	Success, Failed, Skipped
ErrorMessage	TEXT, NULLABLE	Error details if processing failed

Attribute	Type	Description
LinkedRecordType	VARCHAR, NULLABLE	Which entity was created/updated (Sale, Inventory, etc.)
LinkedRecordID	INT, NULLABLE	ID of the created/updated record

6. Dashboard & Visualization Requirements

The application should feature an interactive dashboard as the main landing page, with dedicated sections for different aspects of the business.

6.1 Main Dashboard

Widget	Type	Description
Total Profit (PHP)	KPI Card	Net profit from all sneaker sales (bank transfers minus costs minus fees)
Total Revenue (PHP)	KPI Card	Sum of all bank transfers received
Total Expenses (PHP)	KPI Card	Sum of all expenses (purchases + fees + subscriptions + personal)
Active Inventory Count	KPI Card	Number of items with Status = Available
Inventory Value (PHP)	KPI Card	Total PurchaseCost of all Available items
Profit Over Time	Line Chart	Monthly/weekly profit trend (configurable time range)
Sales by Status	Donut/Pie Chart	Breakdown of current sales by lifecycle status
Action Items	Alert List	Pending confirmations, upcoming shipment deadlines, unreconciled transfers, Attention Needed orders
Recent Activity	Activity Feed	Latest 10-15 email-triggered events with timestamps

6.2 Sales Dashboard

- Sales table with sortable/filterable columns
- Sell-through rate: (Completed Sales / Total Items Purchased) over time
- Average days from purchase to sale
- Average profit margin per sale
- Top performing shoes by profit

6.3 Inventory Dashboard

- Inventory table with search, sort, and filter
- Inventory value by brand/model
- Aging analysis: items in stock longest (potential dead stock)
- Unmatched sales indicator

6.4 Financial Overview

- Income vs. Expenses over time (bar or stacked area chart)
- Expense breakdown by category (pie chart)
- Monthly subscription burn rate
- Bank transfer history and reconciliation status
- Net worth / total funds trend line

7. Notification & Alert System

The application should surface important action items proactively:

Alert Type	Trigger Condition	Urgency
Pending Confirmation	Sale detected; not confirmed within 12 hours	High (24-hour auto-cancel deadline)
Shipment Deadline	Confirmed order within 24 hours of shipping deadline	Critical
Overdue Shipment	Shipment deadline has passed	Critical
Attention Needed	Authentication issue detected	High (48-hour auto-discount timer)
Unreconciled Transfer	Bank transfer received; not matched to any sale after 48 hours	Medium
Unmatched Sale	Sale created with no inventory match after 7 days	Low
Consignment Expiry	Consigned item approaching 90-day storage fee threshold	Medium

Alerts should appear in-app on the dashboard Action Items widget. Future enhancement: optional email digest or push notifications.

8. Technology Stack

Layer	Technology	Purpose
Frontend	React	Dashboard UI, charts, inventory management, forms
Visualization	Recharts / Chart.js	Interactive charts and graphs
UI Components	Tailwind CSS / shadcn/ui	Consistent, responsive design system

Layer	Technology	Purpose
Backend	Flask (Python) or Node.js/Express	REST API, business logic, email parsing
Email Integration	Gmail API (OAuth 2.0)	Email monitoring, parsing, and classification
Database	PostgreSQL / Supabase	Relational data storage with JSON support
Currency API	exchangeratesapi.io or Open Exchange Rates	Real-time USD to PHP conversion
Hosting	TBD (Vercel, Railway, or self-hosted)	Deployment platform

9. Edge Cases & Error Handling

Scenario	Expected Behavior
Sale email arrives for SKU not in inventory	Create sale record; flag as Unmatched. Do not block. Allow manual linking later.
Duplicate email processed	Check EmailProcessingLog by GmailMessageID; skip if already processed.
Bank transfer covers multiple sales	Use BankTransferAllocations table to split across multiple SaleIDs.
Bank transfer amount doesn't match expected USD conversion	Flag for manual review; do not auto-reject. Conversion rates fluctuate.
Return/cancellation for an already-reconciled sale	Reverse the sale status; flag the bank transfer allocation for review.
Email parsing fails	Log the failure in EmailProcessingLog with error details; surface in dashboard.
Gmail API token expires	Auto-refresh using stored refresh token; alert user if refresh fails.
Multiple items with same SKU + Size in inventory	FIFO: select earliest DatePurchased. Let user override if needed.
Order goes Completed then later gets a return	Update sale to Returned; restore inventory; create fee expense if applicable.
Consigned item sells on Alias	New sale email arrives with new order#; match to consigned inventory item.
Cancellation email with ambiguous fee language	Default to No Fee if exact pattern "you will be charged" is not found. Flag for manual review.
Repeated cancellations by seller	Track cancellation count per rolling period; surface as a seller health warning on the dashboard.

10. Future Enhancements (Out of Scope for v1)

- Multi-platform support: StockX, GOAT, eBay integration
- Mobile app (React Native) for on-the-go tracking
- Push notifications for critical alerts
- AI-powered pricing suggestions based on market data
- Shoe data API integration (StockX, GOAT) for auto-populating shoe details and market values
- Tax reporting and exportable financial summaries
- Multi-user support for team-based reselling operations
- Barcode/SKU scanning for quick inventory entry

End of Specification